18. Two straight lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$ intersect each other if $(\vec{c} - \vec{a}) \cdot (\vec{b} \times \vec{d}) = 0$

19. The shortest distance between the two parallel lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{b}$ is $d = \dfrac{|(\vec{c} - \vec{a}) \times \vec{b}|}{|\vec{b}|}$, where $|\vec{b}| \neq 0$

20. If two lines $\dfrac{x - x_1}{b_1} = \dfrac{y - y_1}{b_2} = \dfrac{z - z_1}{b_3}$ and $\dfrac{x - x_2}{d_1} = \dfrac{y - y_2}{d_2} = \dfrac{z - z_2}{d_3}$ intersect, then

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$$

21. A straight line which is perpendicular to a plane is called a normal to the plane.

22. The equation of the plane at a distance p from the origin and perpendicular to the unit normal vector $\hat{d}$ is $\vec{r} \cdot \hat{d} = p$ (normal form)

23. Cartesian equation of the plane in normal form is $lx + my + nz = p$

24. Vector form of the equation of a plane passing through a point with position vector $\vec{a}$ and perpendicular to $\vec{n}$ is $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$.

25. Cartesian equation of a plane normal to a vector with direction ratios a,b,c and passing through a given point (x_1, y_1, z_1) is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$.

26. Intercept form of the equation of the plane $\vec{r} \cdot \vec{n} = q$, having intercepts a,b,c on the x, y, z axes respectively is $\dfrac{x}{a} + \dfrac{y}{b} + \dfrac{z}{c} = 1$.

27. Parametric form of vector equation of the plane passing through three given non-collinear points is $\vec{r} = \vec{a} + s(\vec{b} - \vec{a}) + t(\vec{c} - \vec{a})$

28. Cartesian equation of the plane passing through three non-collinear points is

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0.$$

29. A straight will lie on a plane if every point on the line, lie in the plane and the normal to the plane is perpendicular to the line.

30. The two given non-parallel lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$ are coplanar if $(\vec{c} - \vec{a}) \cdot (\vec{b} \times \vec{d}) = 0$.

31. Two lines $\dfrac{x - x_1}{b_1} = \dfrac{y - y_1}{b_2} = \dfrac{z - z_1}{b_3}$ and $\dfrac{x - x_2}{d_1} = \dfrac{y - y_2}{d_2} = \dfrac{z - z_2}{d_3}$ are coplanar if

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$$

32. Non-parametric form of vector equation of the plane containing the two coplanar lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$ is $(\vec{r} - \vec{a}) \cdot (\vec{b} \times \vec{d}) = 0$ or $(\vec{r} - \vec{c}) \cdot (\vec{b} \times \vec{d}) = 0$.

33. The acute angle θ between the two planes $\vec{r} \cdot \vec{n}_1 = p_1$ and $\vec{r} \cdot \vec{n}_2 = p_2$ is $\theta = \cos^{-1}\left(\dfrac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}\right)$

Chapter 6 Vector Algebra.indd 280 31-01-2020 17:54:10